WORKED SOLUTION

STEP I 1991, Question 14

Probability and Statistics • Counting Triangles

STEP I 1991, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q14

The key idea

Three lengths $a > b > c$ make a non-degenerate triangle exactly when the longest is *strictly* shorter than the sum of the other two, $b + c > a$; the other two triangle inequalities hold automatically. So with a fixed we count, for each b, the values of c that squeeze between $a - b$ and b. These counts go up in steps of 2, which is where the sums of odd (or even) numbers come from. Adding over every possible a then needs only the standard formula for $\sum k^2$.

Solution

When does a triangle exist?

Suppose $a > b > c > 0$. The three triangle inequalities for a non-degenerate triangle are

$$b + c > a, \qquad a + c > b, \qquad a + b > c.$$

The last two hold automatically, because a is already bigger than b and than c. So the three rods form a non-degenerate triangle if and only if $b + c > a$. (If $b + c = a$ the "triangle" is flat, with zero area, and is excluded.)

The largest rod has even length

Fix a. We need integers b, c with

$$a - b < c < b, \qquad b < a.$$

For a given b, the integers c with $a - b + 1 \leq c \leq b - 1$ number

$$(b - 1) - (a - b + 1) + 1 = 2b - a - 1,$$

provided this is positive; otherwise there are none. (Note that $c \geq a - b + 1 \geq 2$ here, so c is automatically a genuine rod length.)

Now let a be even. Then $2b-a-1$ is odd, and it is at least 1 exactly when $b \geq \frac{a}{2}+1$. So b runs from $\frac{a}{2}+1$ up to $a-1$, giving

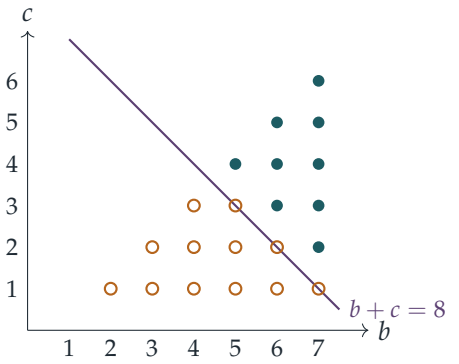
$$\begin{aligned} b = \tfrac{a}{2}+1: &\quad 2b-a-1 = 1, \\ b = \tfrac{a}{2}+2: &\quad 2b-a-1 = 3, \\ &\vdots \\ b = a-1: &\quad 2b-a-1 = a-3. \end{aligned}$$

Each increase of b by 1 adds 2 to the count, so the number of triangles with this largest rod is

$$1+3+5+\cdots+(a-3),$$

as required. (For $a=2$ the sum is empty, which is right: no triangle has longest side 2.) The sum of the first k odd numbers is k^2, and here there are $\frac{a-2}{2}$ terms, so this count is $(\frac{a-2}{2})^2$.

The picture below shows the case $a=8$: each dot is a pair (b,c) with $8>b>c$, and the filled dots are those above the line $b+c=8$. The columns contain 1, 3 and 5 filled dots.



The largest rod has odd length

Now let a be odd. Then $2b-a-1$ is even, and it is positive exactly when $b \geq \frac{a+3}{2}$. As b runs from $\frac{a+3}{2}$ to $a-1$ the counts are $2,4,\ldots,a-3$, so the number of triangles is

$$2+4+6+\cdots+(a-3) = 2\left(1+2+\cdots+\tfrac{a-3}{2}\right) = \frac{a-3}{2}\cdot\frac{a-1}{2} = \frac{(a-1)(a-3)}{4}.$$

(Again $a=3$ correctly gives 0.)

The total number of triangles

The largest rod can be any of $3,4,\ldots,2N+1$ (smaller values give zero anyway).

Even $a=2m$, $m=1,\ldots,N$: the count is $(m-1)^2$, so these contribute

$$\sum_{m=1}^{N}(m-1)^2 = \sum_{k=1}^{N-1}k^2 = \frac{(N-1)N(2N-1)}{6}.$$

Odd $a = 2m+1$, $m = 1, \ldots, N$: the count is $\frac{(2m)(2m-2)}{4} = m(m-1)$, so these contribute

$$\sum_{m=1}^{N}(m^2 - m) = \frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)}{2}$$
$$= \frac{N(N+1)(2N-2)}{6} = \frac{(N-1)N(N+1)}{3}.$$

Adding, the number of selections that form a non-degenerate triangle is

$$T_N = \frac{N(N-1)}{6}\big[(2N-1) + 2(N+1)\big] = \frac{N(N-1)(4N+1)}{6}.$$

Ask yourself Why split the sum by parity of a rather than summing a single formula?

The count for a given a is $\frac{(a-2)^2}{4}$ when a is even but $\frac{(a-1)(a-3)}{4} = \frac{(a-2)^2-1}{4}$ when a is odd. These are different polynomials in a, so writing $a = 2m$ or $a = 2m+1$ turns each into a simple polynomial in m that we can sum with the standard formulae.

Verifying for $N = 3$

The formula gives $T_3 = \frac{3\cdot 2\cdot 13}{6} = 13$. Listing directly with rods $1, \ldots, 7$, writing triangles as (a, b, c):

- $a = 4$ (one): $(4,3,2)$;
- $a = 5$ (two): $(5,4,2)$, $(5,4,3)$;
- $a = 6$ (four): $(6,4,3)$, $(6,5,2)$, $(6,5,3)$, $(6,5,4)$;
- $a = 7$ (six): $(7,5,3)$, $(7,5,4)$, $(7,6,2)$, $(7,6,3)$, $(7,6,4)$, $(7,6,5)$.

That is $1 + 2 + 4 + 6 = 13$ triangles, in agreement with the formula (and with $1 = 1^2$, $4 = 2^2$ for even a and $2 = 2\cdot 1$, $6 = 3\cdot 2$ for odd a).

The probability

Drawing three rods at random without replacement, every one of the $\binom{2N+1}{3}$ unordered selections is equally likely. Since

$$\binom{2N+1}{3} = \frac{(2N+1)(2N)(2N-1)}{6},$$

the probability of a non-degenerate triangle is

$$\frac{T_N}{\binom{2N+1}{3}} = \frac{N(N-1)(4N+1)}{(2N+1)\,2N\,(2N-1)} = \frac{(N-1)(4N+1)}{2(4N^2-1)},$$

cancelling the factor N and using $(2N+1)(2N-1) = 4N^2 - 1$. This is the required result.

Checks

- For $N = 3$ the probability is $\frac{2 \cdot 13}{2 \cdot 35} = \frac{13}{35}$, matching 13 triangles out of $\binom{7}{3} = 35$ selections.
- As $N \to \infty$ the probability tends to $\frac{4N^2}{8N^2} = \frac{1}{2}$. This matches the continuous version: for three independent uniform lengths on $[0, 1]$, the chance that the largest is less than the sum of the other two is $\frac{1}{2}$.
- A brute-force count over all triples, for every N from 2 to 29, agreed exactly with T_N and with the probability formula.

Common slips

- Using $b + c \geq a$: equality gives a flat triangle, which the question excludes, and every count comes out too big.
- Checking all three triangle inequalities separately, or forgetting that only $b + c > a$ matters once a is the largest.
- Getting the range of b off by one; test it on a small case such as $a = 6$ (triangles $(6, 4, 3)$, $(6, 5, 2)$, $(6, 5, 3)$, $(6, 5, 4)$, i.e. $1 + 3$).

Beyond STEP

Counting lattice points in a region, as in the picture, is a discrete version of computing an area, and as N grows the count divided by $\binom{2N+1}{3}$ approaches a volume ratio in the unit cube. This passage from discrete to continuous uniform distributions, and "geometric probability" questions like the broken-stick problem, are classic first-year topics. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read more.

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